

SECTION 35 20 23

DREDGING

08/20

PART 1 GENERAL

1.1 GENERAL INFORMATION

This Section covers furnishing suitable dredging plant and performing all work required to remove the specified materials from within the prescribed work area limits as indicated, and placement of the dredge material within the prescribed material placement areas or at a Contractor-furnished, State-permitted, upland placement site(s). The Contractor is responsible for making their own investigation of submerged, surface, and overhead structures in the work areas and other locations they find necessary to traverse. The exact location, depths, and heights of various structures including, but not limited to submarine cables, pipes, highlines, docks, piers, bulkheads, and bridges (as applicable), are not known and it will be necessary for the Contractor to ascertain interference problems and notify the respective owners in advance of dredging operations. The Contractor is responsible for making necessary arrangements with the respective owners of the structure(s) to assure satisfactory completion of dredging in the vicinity with a minimum interruption of service, and must perform their operations in such a manner as will avoid damage to these facilities.

1.2 DEFINITIONS

1.2.1 Hard Material

Hard material is defined as material requiring [blasting or] [the use of] special equipment for economical removal, and includes boulders or fragments too large to be removed in one piece by the dredge.

1.2.2 Container and Transport Vessels

The term "container vessel" refers to any vessel used to contain, or hold, dredged material (scow, barge, hopper dredge, etc.). The term "transport vessel" refers to any vessel (tugboat, hopper dredge, work boat, etc.) used to move a container vessel. In the case of a hopper dredge, the hopper dredge may act as both a "container vessel" and a "transport vessel". If hopper dredge engines are not used for propulsion, and another vessel (tugboat, for example) is used to move it, then the hopper dredge would be the "container vessel" and the tugboat would be the "transport vessel".

Vessel Identification and Certification: All vessels (container and transport) that will be used, and/or may be used, during this contract must be identified prior to the first day of dredging. Vessels identified for use on the contract must be certified to be fully operational, mechanically sound, completely seaworthy, and free of leaks or other defects.

1.2.3 Environmental Bucket

An environmental bucket is required this project. An environmental bucket is required for dredging silt or other fine-grained unconsolidated materials and/or sediments classified as Class B or Class C according to

NYSDEC In-Water and Riparian Management of Sediment and dredged Material
Technical & Operational Guidance Series (TOGS) 5.1.9.

- a. The bucket must be constructed with sealing gaskets or an overlapping sealed design at the jaws, and seals or flaps positioned at locations of vent to minimize the loss of material during transport through the water column and into the barge.
- b. Any seals or flaps designed and/or installed at the jaws and locations of vent openings must tightly cover these openings while the bucket is lifted through the water column and into the barge. If excessive loss of water and/or sediments from the bucket is observed from the time of its breaking the water surface to crossing the barge gunwale, the Permittee must halt dredging operations and inspect the bucket for defects. Operations must be suspended until all necessary repairs or replacements are made.
- c. Bucket hoist speed must be limited to approximately 2 feet per second. The bucket must be lifted in a continuous motion through the water column and into the barge.
- d. The bucket must be lowered to the level of the barge gunwales prior to the release of load.
- e. Where the environmental bucket repeatedly encounters resistance from consolidated materials (e.g., stone, rock), a conventional digging bucket may be used. The Department must be provided with written notice of intent to switch buckets at least 48 hours (2 business days) prior to use of the conventional bucket.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Soundings or Sweepings; G

1.4 METHOD OF COMMUNICATION

Provide a system of communication between the dredge crew and the crew at the placement area. A portable two-way radio is acceptable.

1.5 ENVIRONMENTAL COMPLIANCE AND PROTECTION

Comply with conditions and requirements of State or Federal permits. The Government will secure the permit for dredging and placement of material as indicated. See Section 00 90 02.

During the life of the contract, provide and maintain environmental protective measures. Also, environmental protective measures required to correct conditions, such as oil spills or debris, that occur during the dredging operations, must be provided. Comply with Federal, State, and

local regulations pertaining to water, air, and noise pollution.

1.6 BASIS FOR BIDS

Refer to Division 00 and Division 01.

PART 2 PRODUCTS

2.1 CHARACTER OF MATERIAL

Sampling and testing was conducted by the Government in 2024. The results are included in the specifications in Section 00 90 01 as Attachment K.

PART 3 EXECUTION

3.1 INSPECTION

Inspect the work, keep records of work performed, and ensure that gages, targets, ranges, and other markers are in place and usable for the intended purpose. Provide, at the request of the Contracting Officer, boats, boatmen, laborers, and materials necessary for inspecting, supervising, and surveying the work. When required, provide transportation for the Contracting Officer and inspectors to and from the placement area and between the dredging plant and adjacent points on shore.

The dredge plant will be inspected by the Contracting Officer, or their representative, prior to beginning work to ensure total dredging plant is in safe working condition. Before any machinery or mechanized equipment is placed in service, it must be inspected and tested by the Contractor and certified to be in safe operating condition.

3.2 DREDGING

3.2.1 Order of Work

The Order of Work shall be determined by the Contractor and discussed during the Preconstruction Conference. The Government reserves the right to change the order of work at any time.

3.2.2 Interference with Navigation

Minimize interference with the use of channels and passages. The Contractor is responsible for shifting or moving of dredges or the interruption of dredging operations to accommodate the movement of vessels and floating equipment, if necessary. Adhere to Coast Guard Regulations for passing vessels.

3.2.2.1 Compensation for Interruption of Operations

If dredging operations are interrupted due to the movement of vessels or floating equipment, no adjustment in the contract time for completion will be made as provided by the contract. The Contractor is required to document the loss of dredging time due to vessel movements on the daily report of the dredging QC and Production form.

3.2.3 Lights

Each night, between sunset and sunrise and during periods of restricted visibility, provide lights for floating plants, pipelines, ranges, and markers. Also, provide lights for buoys that could endanger or obstruct navigation. When night work is in progress, maintain lights from sunset to sunrise for the observation of dredging operations. Lighting must conform to United States Coast Guard requirements for visibility and color.

3.2.4 Navigation Warnings

Furnish and maintain navigation warning signs along any pipeline. Provide notice to increase public awareness of potential hazards presented by dredge plant equipment by stating the location, date of construction, equipment mooring, marshaling areas, using local newspapers, radios, television, waterway users associations, or other appropriate area specific communication networks. Ensure that an announcement is made through the same networks at the beginning of the dredging operation. Make periodic updates/status announcements at intervals of not more than a month throughout the contract life.

Display a sturdy and prominent warning sign at all public boat marinas within 10 miles of dredging operations or moored equipment. The Contractor is responsible for keeping this sign current with respect to the dredging operations or equipment.

3.2.5 Ranges, Gages, and Lines

Provide, set, and maintain ranges, buoys, and markers needed to define the work and to facilitate inspection. Establish and maintain gages in locations observable from each part of the work so that the depth may be determined. Suspend dredging when the gages or ranges cannot be seen or followed. The Contracting Officer will furnish, upon request by the Contractor, survey lines, points, and elevations necessary for the setting of ranges, gages, and buoys.

3.2.6 Dredge Plant and Equipment

- a. Maintain all dredge plant and associated equipment such as, but not limited to, scows, coamings, barges, and pipelines, to meet the requirements of the work. Promptly repair leaks or breaks along pipelines.

3.2.7 Layout of Work

The Contractor will be provided layout charts for all dredging areas based on the schedule of work, the Contractor's hydrosurvey requests, and/or the results of Government hydrographic survey monitoring. The most current Government hydrosurvey information will be displayed on the dredging layout charts and provided to the Contractor. Other pertinent information to be included with each layout package will be: payment method, required and maximum dredging depths, coordinates for the material placement site, distance one-way from dredging location to placement site, specific placement site instructions, supplemental tide gauge information, horizontal control locations, and computed hydrosurvey quantities.

3.2.7.1 Quantity of Material

The total estimated amount of material to be removed from within the

specified limits are shown on Section 00 01 00 "Solicitation, Offer and Award Form" and on Section 01 30 00 "ADMINISTRATIVE REQUIREMENTS", Paragraph 1.9 "Estimated Quantities". The quantities listed are estimates only.

The estimated quantities for bidding purposes and for application of the "FAR 52.212-11, Variation in Estimated Quantity" shall be 15% higher or lower for the contract Work, including overdepth. The quantities are calculated based on the contour dredging method where payment is allowed for all material up to Overdepth Grade except in areas below project grade or on side slopes. The Contractor shall within the limits of available funds, complete the work specified whether the quantities involved are greater or less than those estimated.

3.2.7.2 Overdepth Dredging

To cover unavoidable inaccuracies of dredging processes, material actually removed to a depth of [CLIN-BASED ALLOWABLE OVERDEPTHS AS STATED IN BID SCHEDULE] ~~two (2) feet~~ below the minimum required depth specified and within the dredging limits will be measured and paid for at full contract price. The quantity of overdepth dredging is based on the contour dredging method where payment is allowed for all material up to Overdepth Grade except for areas below project grade.

3.2.7.3 Side Slopes

Dredging the side slopes shall follow, as closely as practicable, a 1:4 side slope with 2 feet allowable overdepth, see 3.2.7.2 "Overdepth Dredging" for overdepth dredging limits. Dredging on side slopes shall follow, as closely as practicable, the lines indicated or specified. A [CLIN-BASED ALLOWABLE OVERDEPTH AS STATED IN BID SCHEDULE] ~~2-foot~~ allowance will be made for dredging beyond the indicated or specified side slopes, except as provided herein. The amount of material excavated from side slopes will be determined by either cross-sections or computer, or both. The quantity of side slope material dredging is based on the contour dredging method where payment is allowed for all material up to Overdepth Grade except below project grade on side slopes.

Material removed to form or maintain the required side slopes within the basic work template is classified as and paid under the basic/required dredging item(s). This classification applies even where the side slope prism overlaps with areas designated as option(s). Exercising any option dredging item does not reclassify or convert material required for basic side slope achievement to option material.

3.2.8 Obstructions and Debris

Previous maintenance dredging operations in the vicinity of this project have encountered debris such as, but not limited to, metal bands, pallets, pieces of broken cable, rope, fire hose, and broken piles. The Contractor is responsible for the disposal of the removed debris. This disposal must occur outside the limits of government property, and done so in accordance with all federal, state, and municipal regulations.

The Government has no knowledge of existing wrecks, wreckage, or other material of such size or character as to require the use of explosives or special or additional plant for its economical removal.

If actual conditions differ from those stated or shown, or both, an adjustment in contract price or time for completion, or both, will be made in accordance with "FAR 52.236-2, Differing Site Conditions."

3.2.9 Dredging Requirements

a. Dredging Areas

- (1) BASIC WORK 1 Dredging Area ~~1 (Basic Work)~~: requires dredging to the plane of -17.5 feet Mean Lower Low Water (MLLW) plus 02 feet allowable overdepth~~+~~.
- (2) BASIC WORK 2 Dredging Area: requires dredging to the plane of -15 feet MLLSW plus 2 feet allowable overdepth.
- (3) OPTION 1 Dredging Area ~~2 (Option)~~: requires dredging to the plane of -12 feet MLLW plus 2 feet allowable overdepth~~+~~.
- (4) OPTION 2 Dredging Area ~~3 (Option)~~: requires dredging to the plane of -12 feet MLLW plus 2 feet allowable overdepth).

3.2.10 Quality Control

Establish and maintain quality control for operations to assure compliance with contractual requirements and maintain records of this quality control for dredging operations.

While performing all dredging work control the horizontal positioning of the dredge with electronic positioning.

3.2.11 Surveys during Progress of Work

Quality of dredging depth will be determined by soundings or sweepings taken behind the dredge as work progresses. The Contractor will take progress soundings or sweepings. The Contractor will submit all data files and surge information, including mapping of these surveys, to the COR for their review. See Section 01 20 00 for Price and Payment Procedures.

3.2.12 Salvaged Material

Anchors, chains, firearms, and other articles of value, which are brought to the surface during dredging operations, must remain or become the property of the Government and will be placed on shore at a convenient location near the site of the work, as directed.

3.2.13 Safety of Structures

The prosecution of work must ensure the stability of piers, bulkheads, and other structures lying on or adjacent to the site of the work, insofar as structures may be jeopardized by dredging operations. Repair damage resulting from dredging operations is the responsibility of the Contractor, insofar as such damage may be caused by variation in locations or depth of dredging, or both, from that indicated or permitted under the contract. The Contractor is responsible for coordinating with the owner of the structure for any necessary repairs.

3.3 PLACEMENT OPERATIONS

3.3.1 Placement of Excavated Materials

Provide for safe transportation and placement of dredged materials. The placement of dredged materials in unauthorized places is forbidden. Comply with rules and regulations of all, federal, state and local authorities.

3.3.2 Charges for Material Placement

The Contractor is responsible for payment of charges imposed by the Contractor-furnished, State-permitted placement site(s).

3.3.3 Misplaced Dredged Material

Any dredged materials deposited at locations other than in areas designated or approved by the Contracting Officer will be considered misplaced material and will not be paid for until the Contractor, at his own expense, removes and deposits such misplaced material where directed. This required removal and redeposit of the misplaced material and any necessary placement site restoration work is not the basis for a time extension or additional compensation under this contract.

The Contractor will immediately notify the Contracting Officer or his Representative if any material is misplaced or when infill or shoaling beyond the Contractor's control occurs.

3.4 MEASUREMENT

The government will perform a before-dredge (BD) survey of the entire work area prior to the commencement of work. Once the Contractor completes the required work and the contractor has submitted a complete survey of the entire dredge area along with all associated files, the government will perform an after-dredge (AD) survey of the same area. The material removed between these two surveys within the allowable footprint and up to the maximum dredge template will be computed. This computation will be made using the TIN method.

Final quantities will be subject to deductions or correction of deductions previously made because of excessive overdepth, dredging outside or authorized areas, or placement of material in an unauthorized manner.

3.4.1 Method of Measurement

Quantity of material removed that will be paid for will be measured by **cubic yard** by means of volume difference from soundings taken before and after dredging. The drawings represent existing conditions based on current available information, but will be verified and corrected, if necessary, by soundings taken before dredging in each locality. The Contractor has the option of being present when such soundings are made.

3.4.2 Monthly Estimates

Monthly estimates of work completed will be based on the result of soundings taken by the Contractor during the progress of the work. Deductions will be made for dredging and placement not in accordance with the specifications.

3.5 FINAL EXAMINATION AND ACCEPTANCE

As soon as practicable after the completion of areas, which in the opinion of the Contracting Officer, will not be affected by further dredging operations, each area will be examined by the Government by sounding or sweeping, or both. Remove shoals and lumps as required by methods approved by the Government. However, if the bottom is soft and the shoal areas form no material obstruction to navigation, removal may be waived at the discretion of the Government. The Contractor will be notified when soundings or sweepings are to be made and will be permitted to accompany the sounding or sweeping party and to inspect the data and methods used in preparing the final quantity for payment. When areas are found to be in a satisfactory condition, the work therein will be accepted as complete.

Re-dredging at the Contract price, within the limit of available funds, may be completed with the consent of both the Government and the Contractor when infill or shoaling beyond the Contractor's control occurs in any area previously accepted.

This contract has no partial acceptance areas. Any survey related disagreement(s) as it pertains to this contract must be submitted in writing to the Contracting Officer or his representative within 5 business days of occurrence of such survey related event. Failure to notify the Contracting Officer or his representative within 5 business days of such survey related event disqualifies the Contractor from any such related claims.

3.6 PLANT REMOVAL

Upon completion of the work, remove all dredging plant, including ranges, buoys, piles, and other markers or obstructions.

-- End of Section --